ECOVISION ACADEMY

MACHINE LEARNING and DATA SCIENCE with PYTHON

Duration: **30 Hours**

Course Fees: **₹449**

▸ Level: Beginner to Advanced

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SECTION 1: Introduction to Data Science & Python Foundations

- Course Overview and Roadmap
- What is Data Science?
- Introducing Data Science and Python
- Python Data Science Ecosystem
- Essential Python Packages
- Installing Anaconda
- Starting Jupyter Notebook
- Basics of Jupyter Notebook
- Markdown Syntax

SECTION 2: NumPy for Numerical Computing

- Creating NumPy Arrays
- One-Dimensional Arrays
- Two-Dimensional Arrays
- NumPy Fast Operations
- Functions in NumPy
- Random Numbers and Distributions
- Creating DataFrames
- Reading Data Files
- Subsetting DataFrames
- Boolean Indexing
- Summarizing Data
- Grouping and Aggregation

SECTION 3: Data Handling with Pandas

- Creating DataFrames
- Reading Data Files
- Subsetting DataFrames
- Boolean Indexing
- Grouping and Aggregation
- Creating Data Files
- Subsetting DataFrames
- Boolean Indexing
- Summarizing Data
- Loading Data into Python

SECTION 4: Exploratory Data Analysis & Visualization

- Introduction to EDA
- Exploratory Analysis of a Dataset
- Introduction to Matplotlib
- Graphs with Matplotlib
- Graphs with Seaborn
- Graphs with Pandas
- Introduction to EDA
- Exploratory Analysis of a Dataset
- Introduction to Matplotlib
- Graphs with Matplotlib
- Graphs with Pandas
- Outlier Detection and Treatment

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SECTION 5: Data Preprocessing & Feature Engineering

- Data Cleaning
- Handling Missing Values
- Categorical Data
- Text Data
- Feature Engineering
- Feature Selection
- Dimensionality Reduction
- Preparing Data for Machine Learning

SECTION 6: Introduction to Machine Learning

- Introduction to Machine Learning
- Types of Machine Learning
- Supervised Learning
- Unsupervised Learning
- Introduction to Scikit-learn
- Toy Datasets
- Machine Learning Workflow
- Train-Test Split

SECTION 7: Supervised Learning Algorithms

- Linear Regression
- Logistic Regression
- K-Nearest Neighbors
- Naive Bayes
- Regression vs Classification

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SECTION 8: Tree-Based & Ensemble Models

- Decision Trees
- Random Forest
- Bias-Variance Tradeoff
- Model Interpretability

SECTION 9: Model Evaluation & Optimization

- Performance Metrics
- Validation Metrics
- Cross-Validation
- Hyperparameter Optimization
- Grid Search
- ML Pipelines
- Testing and Validation

SECTION 10: Advanced Topics & Applications

- Unsupervised Learning Overview
- K-Means Clustering
- Cluster Analysis
- Time Series Analyses

- Tokenization
- Stop-Words Removal
- Text Normalization
- Word Frequencies
- Movie Recommendation System
- Practical Data Science Techniques
- Course Wrap-Up